Renewal Parts **RP01400002E**

Effective April 2009
Supersedes February 2009

Vintage Westinghouse panelboards



PRL1-LX



PRL1 / PRL2



PRL3



CDP, HCDP



FDP

Contents

Description **Page**

Vintage Westinghouse® panelboards

Procedure for identifying renewal parts 2
Distributor ordering instructions 2
PRL1 3
PRL2 3
PRL3 6
PRL1-LX, PRL3-LX 8
CDP, HCDP 10
Breaker cross-reference 13
FDP 16
Trim locks 19
Trim clamps 19
Sumter aftermarket services 20

Powering Business Worldwide

Procedure for identifying renewal parts

1. Identify the type of panelboard (PRL1, PRL2, PRL3, PRL1-LX, PRL3-LX, CPD, FDP) from the data found on the nameplate. Critical information includes:

- Type of panelboard
- Ampere rating of the main
- Ampere rating of the neutral
- Voltage
- Type of service (phase/number of wires)
- General order number

2. Refer to the listing below and turn to the proper section in this renewal parts data to identify standard parts and components.

Description **Page**

PRL1 and PRL2 3
PRL3 6
PRL1-LX, PRL3-LX. 8
CDP, HCDP 10
Breaker cross-reference 13
Series C/Energy Sentinel. 15
FDP 16
Trim locks 19
Trim clamps 19

3. This book identifies those replacement parts most frequently ordered and which are readily available from stock. These parts can be ordered by style or catalog number to speed up processing and delivery.
4. In the event the nameplate is missing, it may still be possible to identify the panel type by the location of the neutral bar. **Figure 1** below shows the position of the neutral in the panelboard.

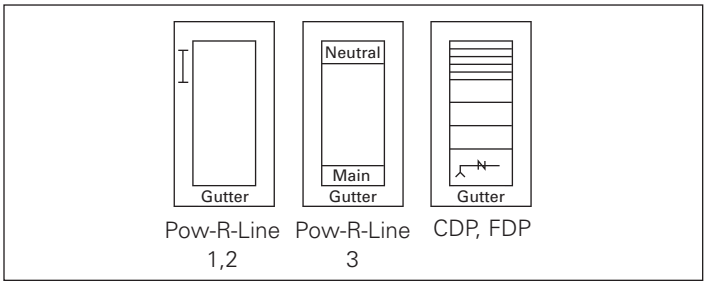


Figure 1. Panelboard neutral locations

Box width may also be used to help identify the panelboard type. Standard width for PRL1, PRL2 and PRL3 is 20.00 inches (508.0 mm). CDP, FDP standard widths are 30.00, 38.00, 42.00, 48.00 and 50.00 inches (762.0, 965.2, 1066.8, 1219.2 and 1270.0 mm).

⚠ CAUTION!

HAZARDOUS VOLTAGE WILL CAUSE SEVERE INJURY OR DEATH. TURN OFF POWER SUPPLY TO EQUIPMENT BEFORE WORKING ON IT.

Distributor ordering instructions

1. Specify the part by catalog number
2. Refer to PL01400002E for pricing information. Discount Symbol CE9 applies
3. Enter the order on Vista, location Q49 or SUM

Note: For CDP retrofit kits shown on **page 12**, you must enter the catalog number, list price, standard multiplier and location Q49 or SUM.

4. When ordering a new circuit breaker to add to a CDP panelboard, the following options are available:

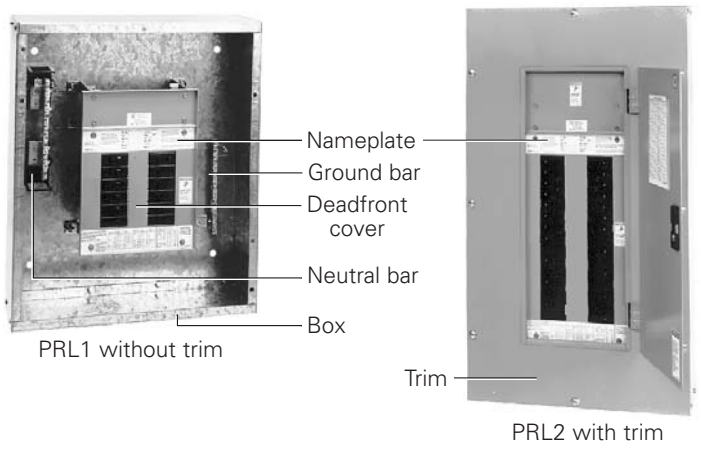
- Take the required breaker from your stock (or order from W34) and order the connector kit (**pages 14 and 18**) from the Sumter Plant

or

- Select the correct breaker retrofit kit (**page 12**) which includes the breaker and connector kit in one package, one shipment and one invoice from the Sumter Plant

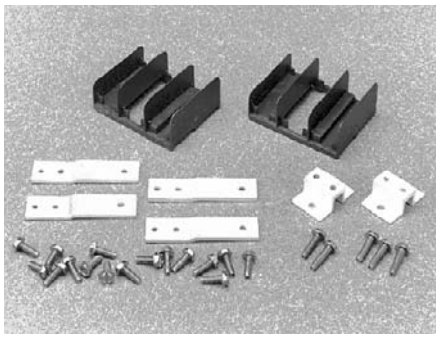
5. Selling policy 25-000 applies
6. If additional support is required, contact the Sumter Plant Aftermarket Center at 1-803-481-6843. A complete listing of available services is listed on **page 20**

PRL1 and PRL2 panelboards



PRL1 and PRL2

Replacement connectors for PRL1 and PRL2



Replacement connectors for PRL1 and PRL2

Each kit contains two sets of A, B, and C phase connectors, phase isolators, hardware and instructions for mounting up to 12 poles. Silver-plated copper connectors are suitable for use with copper or aluminum bus. Maximum amperes connected to any one connector cannot exceed 140A.

Table 1. Replacement connectors

Panel type	Catalog number
PRL1	KPRL1BAB
PRL2	KPRL2GHB

Replacement boxes and trims for PRL1 and PRL2

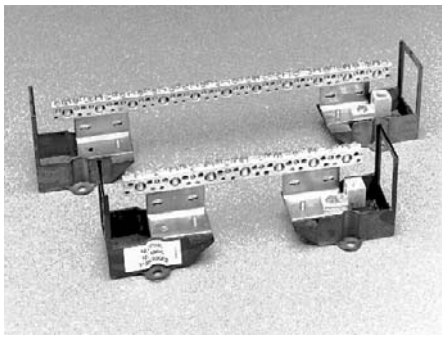
Standard box and trim sizes are shown in **Table 2**. For other sizes, contact your local Satellite. For Type 12/3R enclosures, refer to **page 8**. Trim mounting may be either surface or flush. Flush trims overlap the box by 0.75-inch (19.1 mm) on all sides. Denote on order by using either S or F in the catalog number.

Table 2. Replacement boxes and trims

Box dimensions in inches (mm)			Type 1	
			Catalog number	
Height	Width	Depth	Box	Trim
21.00 (533.4)	20.00 (508.0)	5.75 (146.1)	YS2021	LT2021S/F
24.00 (609.6)			YS2024	LT2024S/F
27.00 (685.8)			YS2027	LT2027S/F
30.00 (762.0)			YS2030	LT2030S/F
33.00 (838.2)			YS2033	LT2033S/F
36.00 (914.4)			YS2036	LT2036S/F
39.00 (990.6)			YS2039	LT2039S/F
42.00 (1066.8)			YS2042	LT2042S/F
45.00 (1143.0)			YS2045	LT2045S/F
48.00 (1219.2)			YS2048	LT2048S/F
51.00 (1295.4)			YS2051	LT2051S/F
54.00 (1371.6)			YS2054	LT2054S/F
57.00 (1447.8)			YS2057	LT2057S/F
60.00 (1524.0)			YS2060	LT2060S/F
63.00 (1600.2)			YS2063	LT2063S/F
66.00 (1676.4)			YS2066	LT2066S/F
69.00 (1752.6)			YS2069	LT2069S/F
72.00 (1828.8)			YS2072	LT2072S/F
75.00 (1905.0)			YS2075	LT2075S/F
78.00 (1981.2)			YS2078	LT2078S/F
81.00 (2057.4)			YS2081	LT2081S/F
84.00 (2133.6)			YS2084	LT2084S/F
87.00 (2209.8)			YS2087	LT2087S/F
90.00 (2286.0)			YS2090	LT2090S/F

PRL1 and PRL2 panelboards

100A neutral bar

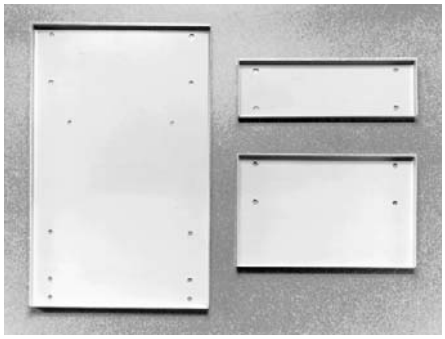


100A neutral bar

Table 3. 100A neutral bar

Description		Style number
A	Boxes 30.00 inches (762.0 mm) and above	5158C03G02
B	Boxes less than 30.00 inches (762.0 mm) high	5158C03G01

Blank covers



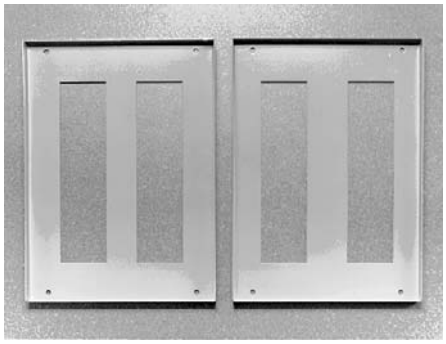
Blank covers

Table 4. Blank covers

Blank covers height in inches (mm)		Style number
A	15.00 (381.0)	3456B99H05
B	3.00 (76.2)	3456B99H01
C	6.00 (152.4)	3456B35H01

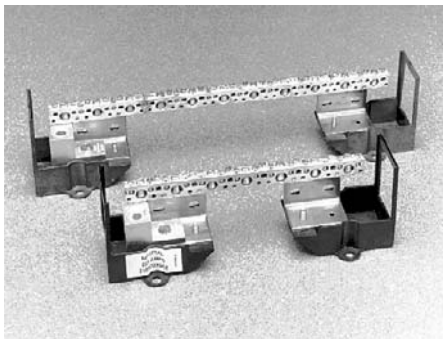
Branch section deadfront covers

PRL1 PRL2



Branch section deadfront covers

225A neutral bar

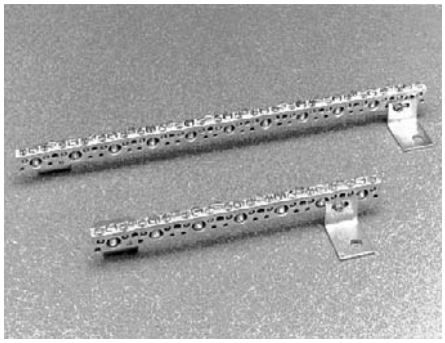


225A neutral bar

Table 5. 225A neutral bar

Description		Style number
A	Boxes 30.00 inches (762.0 mm) and above	5158C03G04
B	Boxes less than 30.00 inches (762.0 mm) high	5158C03G03

Standard ground bar

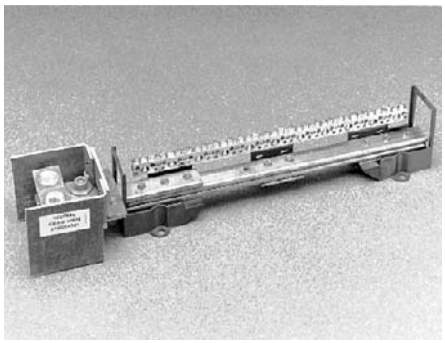


Standard ground bar

Table 6. Standard ground bar

Description		Style number
A	Boxes 30.00 inches (762.0 mm) and above	5158C05G02
B	Boxes less than 30.00 inches (762.0 mm) high	5158C05G01

400/600A neutral bar

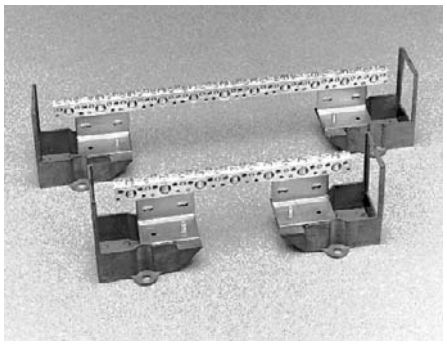


400/600A neutral bar

Table 7. 400/600A neutral bar

Description	Style number
400/600A neutral bar	5158C04G01

Isolated ground bar



Isolated ground bar

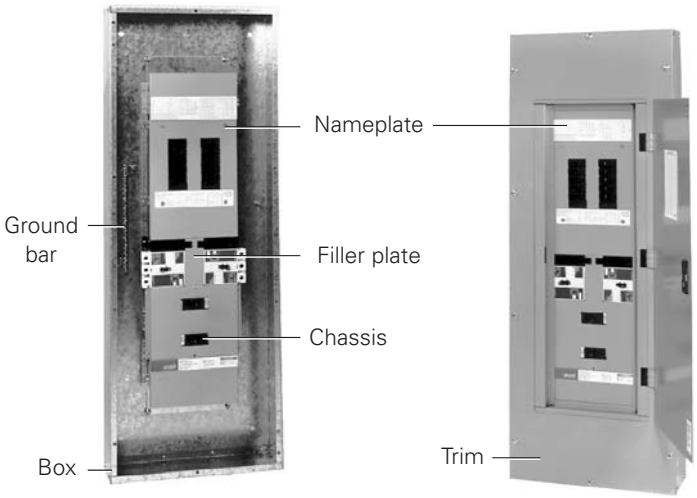
Table 8. Isolated ground bar

Description		Style number
A	Boxes 30.00 inches (762.0 mm) and above	5158C05G06
B	Boxes less than 30.00 inches (762.0 mm) high	5158C05G05

Table 9. PRL1, PRL2 branch section deadfront covers

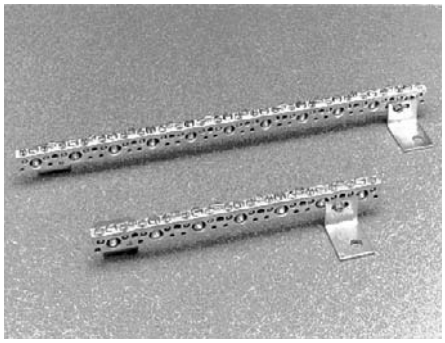
Maximum number of circuits	Cover height in inches (mm)	Cover style number	
		PRL1	PRL2
12	10.00 (254.0)	5156C10H01	5156C09H01
18	13.00 (330.2)	5156C10H02	5156C09H02
24	16.00 (406.4)	5156C10H03	5156C09H03
30	19.00 (482.6)	5156C10H04	5156C09H04
36	22.00 (558.8)	5156C10H05	5156C09H05
42	25.00 (635.0)	5156C10H06	5156C09H06
48	28.00 (711.2)	5156C10H07	5156C09H07
54	31.00 (787.4)	5156C10H08	5156C09H08
60	34.00 (863.6)	5156C10H09	5156C09H09

PRL3 panelboards



PRL3

Standard ground bar



Standard ground bar

Table 10. Standard ground bar

Description		Style number
A	Boxes 30.00 inches (762.0 mm) and above	5158C05G02
B	Boxes less than 30.00 inches (762.0 mm) high	5158C05G01

Replacement boxes and trims for PRL3

Standard boxes and trim sizes are shown in **Table 11**. For other sizes, contact your local Satellite. For Type 12/3R enclosures, refer to **page 8**. Trim mounting may be either surface or flush. Denote on the order by using either S or F in the catalog number. Flush trims overlap the box by 0.75-inch (19.1 mm) on all sides.

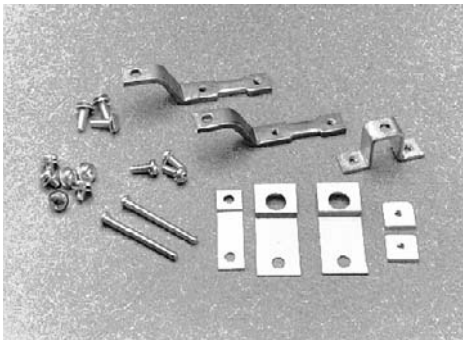
Table 11. Replacement boxes and trims

Box dimensions in inches (mm)			Type 1	
			Catalog number	
Height	Width	Depth	Box	Trim
30.00 (762.0)	20.00 (508.0)	5.75 (146.1)	YS2030	LTT2030S/F
36.00 (914.4)			YS2036	LTT2036S/F
42.00 (1066.8)			YS2042	LTT2042S/F
48.00 (1219.2)			YS2048	LTT2048S/F
54.00 (1371.6)			YS2054	LTT2054S/F
60.00 (1524.0)			YS2060	LTT2060S/F
66.00 (1676.4)			YS2066	LTT2066S/F
72.00 (1828.8)			YS2072	LTT2072S/F
78.00 (1981.2)			YS2078	LTT2078S/F
84.00 (2133.6)			YS2084	LTT2084S/F
90.00 (2286.0)			YS2090	LTT2090S/F

PRL3 panelboards

Replacement connectors for EHD, FDB, FD, HFD, and FDC breakers (240, 480, 600V)

Kit contains A, B and C phase connectors, hardware and instructions to twin mount two 3-pole above listed frame breakers. Maximum amperes per breaker not to exceed 150A, and maximum amperes connected to any one connector not to exceed 200A. (50–150As.)



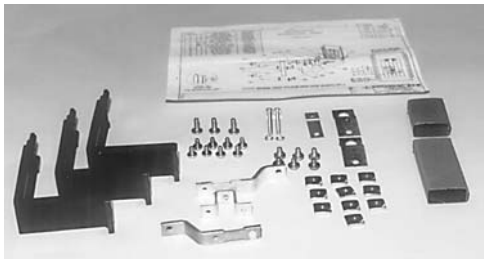
Replacement connectors for EHD, FDB, FD, HFD and FDC breakers

Table 12. Replacement connectors for EHD, FDB, FD, HFD, and FDC breakers

Replacement connectors for breakers	Catalog number
EHD, FDB, FD, HFD, and FDC	KPRL3FD

Replacement connectors for ED, EDH, and EDC breakers

Kit contains A, B, and C phase connectors, hardware cover and instructions to mount one 3-pole above breaker frame. (240V maximum.)



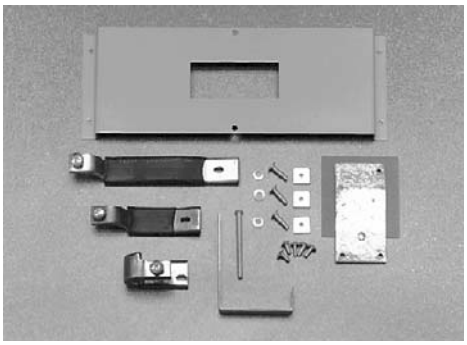
Replacement connectors for ED, EDH, and EDC breakers

Table 13. Replacement connectors for ED, EDH, and EDC breakers

Replacement connectors for breakers	Catalog number
ED, EDH, and EDC	KPRL3ED

Replacement connectors for CA, CAH, and HCA breakers

Kit contains A, B, and C phase connectors, hardware, cover and instructions to mount one 2- or 3-pole CA frame breaker.

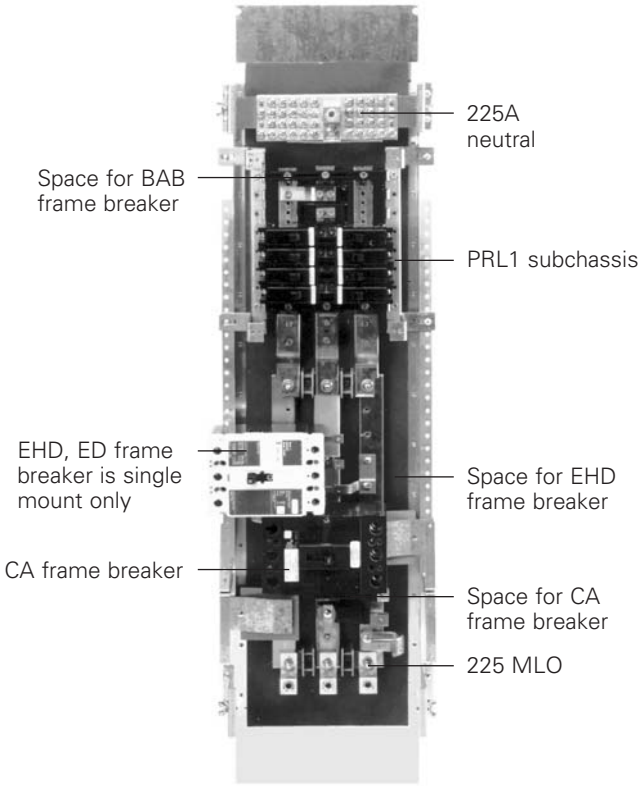


Replacement connectors for CA, CAH, and HCA breakers

Table 14. Replacement connectors for CA, CAH, and HCA breakers

Replacement connectors for breakers	Catalog number
CA, CAH, and HCA	KPRL3CA

Note: The PRL3 connector kits will fit the following panelboards: WEB, WCA, WEHB, WFB, and HWFB.



PRL3 interior

PRL1, 2, 3 special trims and enclosures

Ventilated trim

Required on PRL1 and PRL2, 600A panels only. Order by adding the letter "V" to the standard trim catalog number. ADD 10 percent TO STANDARD TRIM LIST PRICE.

Example: LT2072S becomes LTV2072S.

Refer to **page 3** for the standard trim catalog numbers.



Ventilated trim

Type 12/3R enclosures



Type 12/3R enclosures

Fastrim

Used when concealed trim mounting hardware is required for PRL1, PRL2, and PRL3. Flush trim utilizes Style Number 414A380G01 trim clamps. Surface trim utilizes Style Number 8712C73H01 trim clamps. Trim clamps are included and shipped with the trim. Order by adding the letter "F" to the standard trim catalog number. ADD 20 percent TO STANDARD TRIM LIST PRICE.

Example: LT2072S becomes LTF2072S.

Refer to **pages 3** and **5** for standard trim catalog numbers.



Fastrim

Door-in-door trim

Piano hinge on the right side of the trim provides access to the wiring gutters without requiring removal of the trim. Order by adding the letters "DD" to the standard trim catalog number. ADD 20 percent TO STANDARD TRIM LIST PRICE.

Example: LT2072S becomes LTDD2072S.

Refer to **pages 3** and **5** for standard trim catalog numbers.



Door-in-door trim

The complete enclosure consists of a box and trim. The enclosure meets code requirements for both Type 12 (dust-tight) and Type 3R (rainproof) standards. Features include a laser cut trim with rounded corners, concealed hinges, and a T-handle lock. Gasketing is provided around the trim door. The gasketed box is made of code gauge steel with dripshield and is painted ANSI-61.

Table 15. Type 12/3R enclosures for PRL1, 2, 3

Box dimensions in inches (mm)			Catalog number		
Height	**Width**	**Depth**	**Box**	**PRL1, 2 trim**	**PRL3 trim**
24.00 (609.6)	20.00 (508.0)	6.00 (152.4)	GWPB2024	LWPT2024	—
36.00 (914.4)			GWPB2036	LWPT2036	LWT2036
48.00 (1219.2)			GWPB2048	LWPT2048	LWT2048
60.00 (1524.0)			GWPB2060	LWPT2060	LWT2060
72.00 (1828.8)			GWPB2072	LWPT2072	LWT2072
78.00 (1981.2)			GWPB2078	LWPT2078	LWT2078

Column-type panels

PRL1-LX replacement boxes and trims

- Standard box includes box ends at top and bottom
- Box for trough is supplied with bottom box end only
- Contact the Sumter Plant Aftermarket Center for availability and application information

Table 16. Column-type panels

Box dimensions in inches (mm)			Catalog number		
Height	Width	Depth	Standard box	Box for trough	Surface trim
42.00 (1066.8)	8.63 (219.2)	5.00 (127.0)	114D939G10	114D939G26	6724D78G10
54.00 (1371.6)			114D939G12	114D939G28	6724D78G12
66.00 (1676.4)			114D939G14	114D939G30	6724D78G14
78.00 (1981.2)			114D939G16	114D939G32	6724D78G16

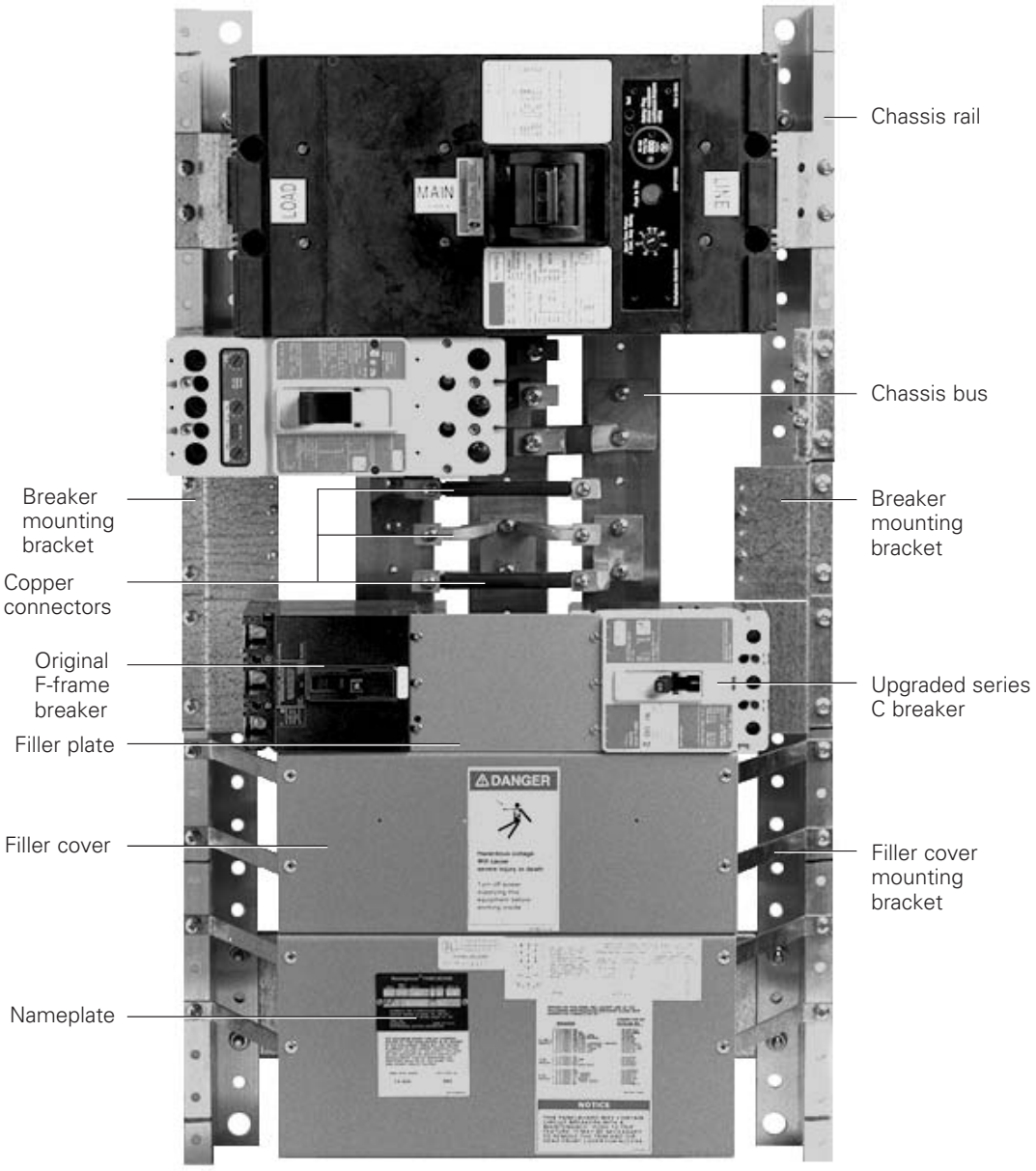
PRL3-LX replacement boxes and trims

- Standard box includes box ends at top and bottom
- Box for trough is supplied with bottom box end only
- Design is for top feed application only
- Contact the Sumter Plant Aftermarket Center for availability and application information

Table 17. PRL3-LX replacement boxes and trims

Box dimensions in inches (mm)			Catalog number		
Height	Width	Depth	Standard box	Box for trough	Surface trim
Main lug only (top feed)					
40.75 (1035.1)	8.63 (219.2)	5.75 (146.1)	6726D69G01	6726D69G07	6726D70G01
49.00 (1244.6)			6726D69G02	6726D69G08	6726D70G02
57.25 (1454.2)			6726D69G03	6726D69G09	6726D70G03
65.50 (1663.7)			6726D69G04	6726D69G10	6726D70G04
73.75 (1873.3)			6726D69G05	6726D69G11	6726D70G05
82.00 (2082.8)			6726D69G06	6726D69G12	6726D70G06
Main breaker (top feed)					
51.75 (1314.5)	8.63 (219.2)	5.75 (146.1)	6726D69G13	6726D69G19	6726D70G07
60.00 (1524.0)			6726D69G14	6726D69G20	6726D70G08
68.25 (1733.6)			6726D69G15	6726D69G21	6726D70G09
76.50 (1443.1)			6726D69G16	6726D69G22	6726D70G10
84.75 (2152.7)			6726D69G17	6726D69G23	6726D70G11
93.00 (2362.2)			6726D69G18	6726D69G24	6726D70G12

CDP/HCDP panelboards



CDP/HCDP panelboard

Branch circuit breaker selection

Step 1: Select the correct breaker.

When selecting a breaker, you need to know the following:

- Amperage
- System voltage
- Available short circuit rating
- Number of poles
- Size and number of wires per phase

Step 2: Use the chart on **page 14** to determine the space required.

Step 3: Create the Retrofit Kit catalog number by following the instructions on **page 12**.

Step 4: Determine if extra filler covers are required. Additional filler covers may be necessary to fill the unused space. Refer to **page 14** for this information.

Note: If the CDP panelboard has a subchassis lighting panel, contact the Sumter Plant Aftermarket Center for breaker selection.

Table 18. Electrical characteristics of branch circuit breakers

Circuit breaker type	Continuous ampere rating	Number of poles	Vac	UL® listed interrupting capacities—kA symmetrical amperes					
				AC rating volts				DC rating volts ①	
				240	277	480	600	125	250
EHD	15–100	1	277	—	14	—	—	10	—
	15–100	2, 3	480	18	—	14	—	—	10
ED	100–225	2, 3	240	65	—	—	—	—	10
EDH	100–225	2, 3	240	100	—	—	—	—	10
EDC	15–225	2, 3	240	200	—	—	—	—	10
FDB	15–225	2, 3	600	18	—	14	14	—	10
FD	15–225	1	277	—	25	—	—	10	—
	15–225	2, 3	600	65	—	25	18	—	10
HFD	15–225	1	277	—	65	—	—	10	—
	15–225	2, 3	600	100	—	65	25	—	22
FDC	15–225	2, 3	600	200	—	100	35	—	22
CA ②	100–225	2, 3	240	10	—	—	—	—	—
CAH ②	100–225	2, 3	240	22	—	—	—	—	—
HCA ②	100–225	2, 3	240	42	—	—	—	—	—
JD, JDB	70–250	2, 3	600	65	—	35	18	—	10
HJD	70–250	2, 3	600	100	—	65	25	—	22
JDC	70–250	2, 3	600	200	—	100	35	—	22
DK	250–400	2, 3	240	65	—	—	—	—	10
KD, KDB	100–400	2, 3	600	65	—	35	25	—	10 ③
HKD	100–400	2, 3	600	100	—	65	35	—	22 ③
KDC	100–400	2, 3	600	200	—	100	50	—	22 ③
LCL	125–400	2, 3	480	200	—	200	—	—	—
LD	300–600	2, 3	600	65	—	35	25	—	22 ③
HLD	300–600	2, 3	600	100	—	65	35	—	25 ③
LDC	300–600	2, 3	600	200	—	100	50	—	25 ③
MDL	300–800	2, 3	600	65	—	50	25	—	22 ③
HMDL	300–800	2, 3	600	100	—	65	35	—	25 ③
ND	600–1200	2, 3	600	65	—	35	25	—	—
HND	600–1200	2, 3	600	100	—	65	35	—	—
NDC	600–1200	2, 3	600	200	—	100	50	—	—
Fused circuit breakers with LFB current limiters									
FDB, LFB	15–150	2,3	600	200	—	200	200	—	—
TRI-PAC integrally fused, current limiting circuit breakers									
FB-P	15–100	2, 3	600	200	—	200	200	—	④
LA-P	70–400	2, 3	600	200	—	200	200	—	④
NB-P	300–800	2, 3	600	200	—	200	200	—	④

① DC ratings apply to substantially non-inductive circuits.

② Not recommended for motor loads.

③ DC ratings not available with electronic trip.

④ 100K based on NEMA® test procedure.

CDP panelboards

CDP breaker retrofit kits

A retrofit kit includes one breaker and all the associated parts required for mounting on a CDP panel chassis. See **Table 19** below.

Table 19. CDP breaker retrofit kits

Frame ampere rating	Breaker frame	Trip ampere range	Mounting type	Space required	
				Inches (mm)	X space
100	EHD	15–60	Twin	4.13 (104.9)	3X
	EHD	70–100	Twin		
	FDB	15–60	Twin		
	FDB	70–100	Twin		
	FD	15–60	Twin		
	FD	70–100	Twin		
	HFD	15–60	Twin		
	HFD	70–100	Twin		
	FDC	15–60	Twin		
	FDC	70–100	Twin		
	FCL	15–100	Twin		
	FB-P	15–100	Twin		
225	FDB	110–225	Twin	4.13 (104.9)	3X
	FD	110–225	Twin		
	HFD	110–225	Twin		
	FDC	110–225	Twin		
	ED	110–225	Twin		
	EDH	110–225	Twin		
	EDC	110–225	Twin		
250	JD	70–225	Twin	4.13 (104.9)	3X
	JD	250	Twin		
	HJD	70–225	Twin		
	HJD	250	Twin		
	JDC	70–225	Twin		
	JDC	250	Twin		
400	KD	100–400	Twin		
	HKD	100–400	Twin		
	KDC	100–400	Twin	5.50 (139.7)	4X
	LCL	125–400	Twin	8.25 (209.6)	6X
	LA-P	70–400	Twin	8.25 (209.6)	6X
600	LD	300–600	Twin	8.25 (209.6)	6X
	HLD	300–600	Twin		
	LDC	300–600	Twin		
800	MDL	400–800	Single	8.25 (209.6)	6X
	HMDL	400–800	Single		
	HND	400–800	Single		
1200	ND	600–1200	Single	8.25 (209.6)	6X
	HND	600–1200	Single		
	NDC	600–1200	Single		

How to create a retrofit kit catalog number

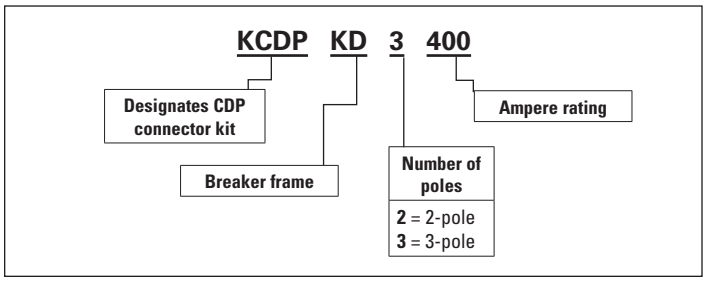
Use "KCDP" prefix and add the catalog number of the specific breaker as shown in **Table 20**.

Twin mounting indicates that only one set of connectors is required to mount two breakers (of similar frames) opposite one another.

A RETROFIT KIT INCLUDES ONE BREAKER ONLY, FOR EITHER SINGLE OR TWIN MOUNTED APPLICATIONS. If a second breaker is required for a twin mounting application, it must be specified as a separate item.

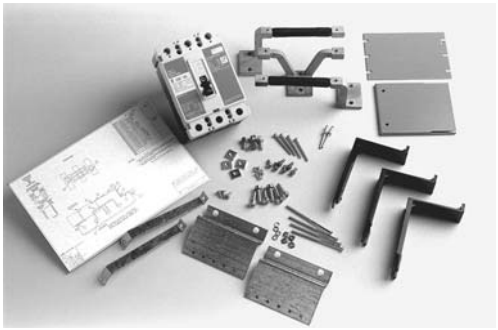
Example: The retrofit kit catalog number for a 3-pole, 400A KD breaker is:

Table 20. Catalog numbering system



CDP retrofit kit

Each kit includes one breaker, copper connectors, mounting brackets, covers, hardware and instructions.



CDP retrofit kit

Breaker cross-reference chart

Use the chart below to cross-reference older industrial type Westinghouse breakers with Series CT breakers. The Series C breakers shown on the right side of the chart use the same "X" space as the older breaker on the left, eliminating the need for any filler covers when retrofitting a CDP panel. By using the retrofit kit catalog number, you can order the breaker, as well as the corresponding CDP connector kit in one easy step.

Note: If the original space was already provided with connectors and hardware to mount the older breakers, then the Series C frames shown here will also fit in that same provision. (The Series C K family of breakers requires an adapter kit called TAD3 when mounting a KD frame breaker onto existing LB frame connectors. This adapter kit is included in orders for either KD retrofit kits or connector kits, but may also be purchased separately for use with existing LB connectors.)

Table 21. Breaker cross-reference chart

Existing breaker		Order Series C retrofit kit	
Breaker frame	**Continuous ampere rating**	**Breaker frame**	**Retrofit kit catalog number**
EB	15–100	EHD	KCDPEHD
EHB	15–100	EHD	KCDPEHD
FB	15–150	FDB	KCDPFDB
HFB	15–150	FD	KCDPFD
KA	70–225	KD	KCDPKD
HKA	70–225	HKD	KCDPHKD
KB	70–250	JD	KCDPJD
HKB	70–250	JD	KCDPJD
JA	70–225	KDB	KCDPKDB
JB	70–250	JDB	KCDPJDB
DA	250–400	DK	KCDPDK
LA	250–600	LD	KCDPLD
HLA	250–600	LD	KCDPLD
LB	70–400	KD	KCDPKD
LBB	125–400	KDB	KCDPKDB
HLB	125–400	KD	KCDPKD
LC	300–600	LD	KCDPLD
HLC	300–600	LD	KCDPLD
MA	125–800	MDL	KCDPMDL
HMA	125–800	MDL	KCDPMDL
MC	400–800	MDL	KCDPMDL
NB	700–1200	ND	KCDPND
HNB	700–1200	ND	KCDPND
NC	600–1200	ND	KCDPND
HNC	600–1200	ND	KCDPND

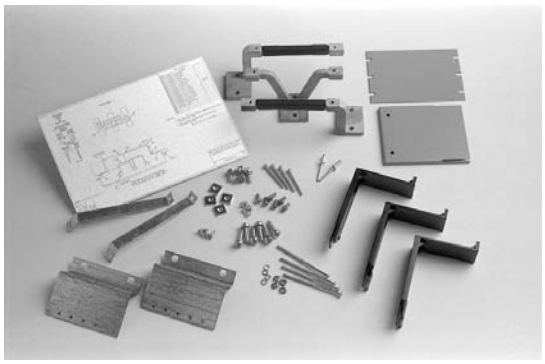
CDP panelboards

CDP breaker connector kits

A connector kit consists of copper connectors, mounting brackets, covers, hardware, and instructions for mounting breaker(s) on a CDP chassis. See below. If breaker is also required, see retrofit kits shown on **page 12**.

Connector kit

Each kit includes copper connectors, mounting brackets, covers, hardware, and instructions.



Connector kit

Table 22. CDP panelboards

Breaker frames	Number of poles	Mounting type	Space required		Connector kit catalog numbers
			Inches (mm)	X space	
EHD, FDB, FD, HFD, FDC ①	3	Twin	4.13 (104.9)	3X	KCDPFB100
FD, FDB, HFD, FDC, ED, EDH, EDC	3	Twin			KCDPED225
FB-P, FCL	3	Twin			KCDPFBP
CA, CAH, HCA	3	Twin			KCDPCA
JD, JDB, HJD, JDC	3	Twin			KCDPKB
DK, KD, KDB, HKD, KDC	3	Twin	5.50 (139.7)	4X	KCDPLB
LA-P	3	Twin	8.25 (209.6)	6X	KCDPLA400
LD, HLD, LDC	3	Twin			KCDPLA600
LCL	3	Single			KCDPLCL
MDL, HMDL	3	Single			KCDPMC
ND, HND	3	Single			KCDPNC ②
NB-P	3	Single			KCDPNBP

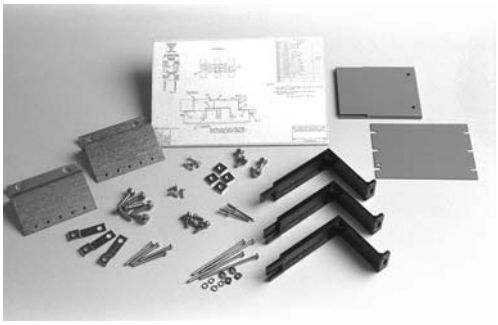
① 100A maximum.

② Connector kits are not rated for 100 percent breakers.

Hardware kit

Each kit includes mounting bracket(s), filler plates and mounting hardware ONLY. Use the appropriate connector kit catalog number and add an "H" to designate hardware only.

Example: KCDPFD100H

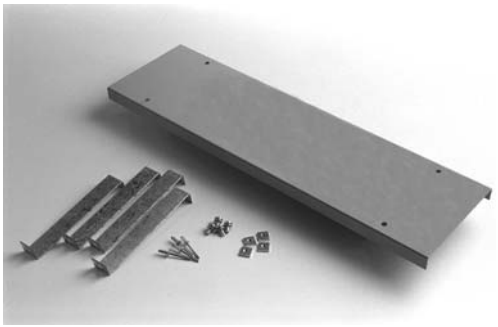


Hardware kit

CDP filler covers

Filler covers are available in any increment of 1X 1.38 inches (35.1 mm). Catalog number is:

KCDP_ X.
Fill in the blank with the size. Included are cover, mounting brackets, and hardware.



CDP filler covers

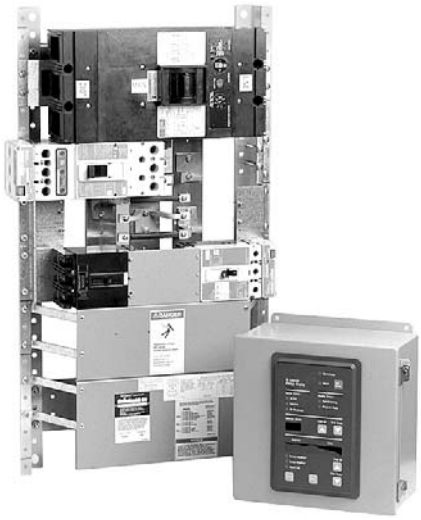
Energy submetering for existing CDP panelboards

The IQ Energy Sentinel is a submetering device that mounts directly on a Series C circuit breaker and can be retrofitted into a CDP panelboard. It monitors both power and energy from 15 to 400A, with an overall accuracy of 99 percent. This high system accuracy is achieved by use of the Cutler-Hammer SURE Plus Chip, which is a sophisticated microprocessor.

All that is necessary to complete an IQ Energy Sentinel installation is to insert it into the load side of a breaker; feed the load conductors through it; connect a neutral wire; and run the shielded twisted pair wire for communications. The IQ Energy Sentinel has a nonvolatile memory, is powered by the circuit breaker, and can be applied on 3-phase, four-wire or 1-phase, three-wire systems.

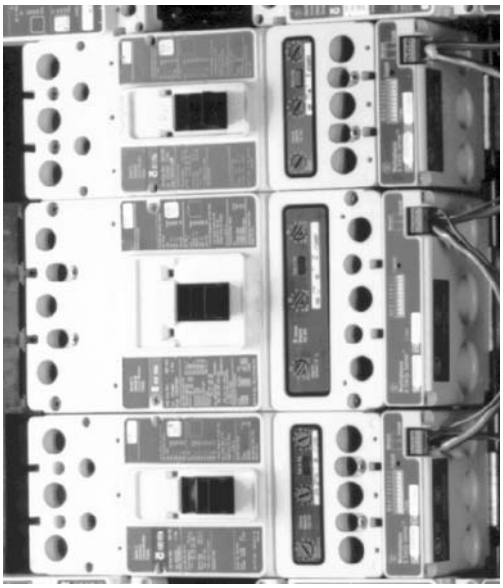
The space-saving design characteristics of the IQ Energy Sentinels means they can be quickly and easily retrofitted onto Series C circuit breakers in existing equipment with no additional space required. Additionally, IQ Energy Sentinels can be installed when upgrading to Series C from older breakers that are physically interchangeable with no additional space requirements.

Power and energy information from IQ Energy Sentinels can be communicated to a personal computer, a panel-mounted central energy display (CED), or even existing building management or distribution control systems.



CDP chassis with energy sentinel

Series C breakers with energy sentinel



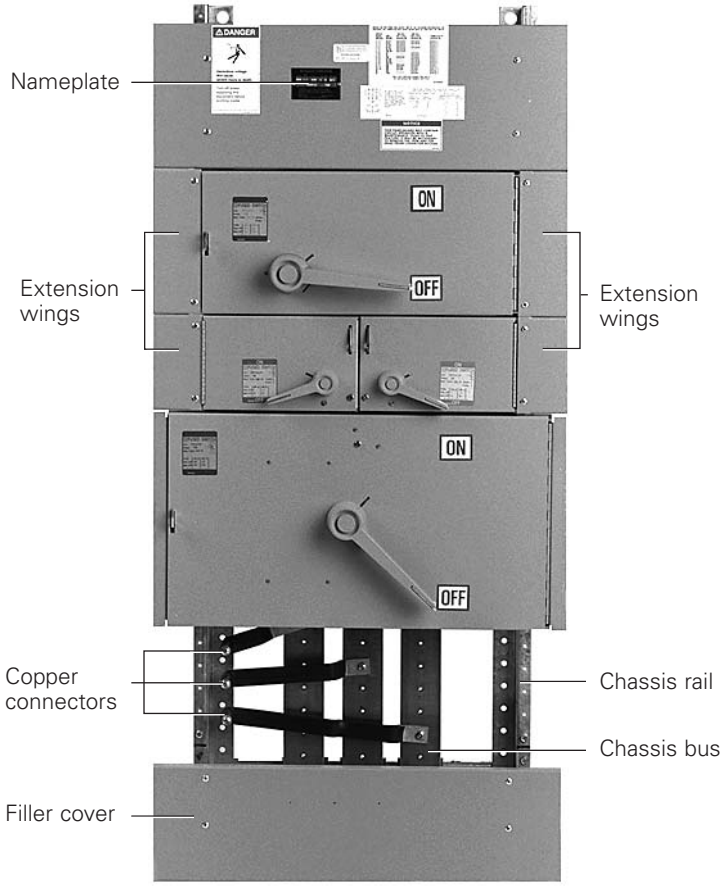
Series C breakers with energy sentinel

Retrofit capabilities

With the advent of this new technology, we can offer the capability to retrofit existing CDP panelboards. This is accomplished by providing a complete system package including:

- Series C breaker retrofit kit
- Energy Sentinel
- All necessary wiring diagrams and mounting instructions
- Start-up assistance
- Communications capabilities with one of the following:
  - IQ Central Energy Display
  - Custom Billing Software Package
  - IMPACC/PowerNet™

FDP panelboards



FDP panelboard

FDP panelboards

Branch FDPW and FDPB

Switch selection

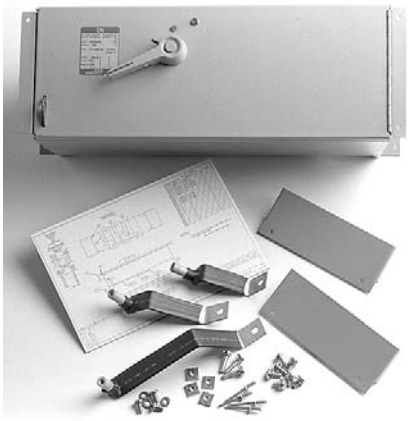
Step 1: Once you know the amperage and system voltage, choose the retrofit kit catalog number from **Table 23**.

Step 2: Check for space requirements from **Table 23**.

Step 3: Determine if extra filler covers are required. Additional filler covers may be necessary to fill the unused space.

Retrofit kit

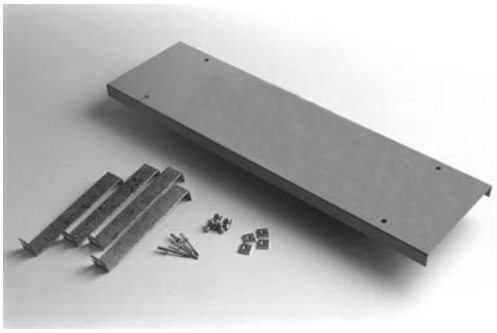
Each kit includes one switch, copper connectors, covers, hardware and instructions.



Retrofit kit

FDP filler covers

Filler covers are available in any increment of 1X 1.38 inches (35.1 mm). Catalog number is: KFDP_ X. Fill in the blank with the size. Included are cover, mounting brackets and hardware.



FDP filler cover

FDPW and FDPB retrofit kits

A retrofit kit includes the switch and all the associated parts required for mounting on a FDP panel chassis.

Switches through 600A are designed for use with rejection type fuses. For other fuse applications, contact the Sumter Plant Aftermarket Center. Switches rated 800 and 1200A are designed for use with Class L fuses. Fuses are not included.

Table 23. 3P retrofit kits

Space required		Switch ampere rating	Catalog number	
Inches (mm)	X space		240V	600V
5.50 (139.7)	4X	30–30 60–60 100–100	KFDPWT3211R KFDPWT3222R KFDPWT3233R	KFDPWT3611R KFDPWT3622R —
6.88 (174.8)	5X	100–100	—	KFDPWT3633R
8.25 (209.6) 12.38 (314.5)	6X 9X	200 400	KFDPBS324R KFDPW325R	KFDPBS364R KFDPW365R
15.13 (384.3) 15.13 (384.3) 20.63 (524.0)	11X 11X 15X	600 800 1200	KFDPW326R KFDPW367 KFDPW368	KFDPW366R KFDPW367 KFDPW368

Switch connector and retrofit kits

FDPW or FDPB switch connector kits and retrofit kits to be installed in vintage Westinghouse FDP panelboards and switchboards. Vertical bus bars must on 3.50 inch (88.9 mm) center-line phase to phase. See **Figure 2**.

Note: If bus bar spacing is 4.25 inches (108.0 mm), these switches will not fit and no parts are available.

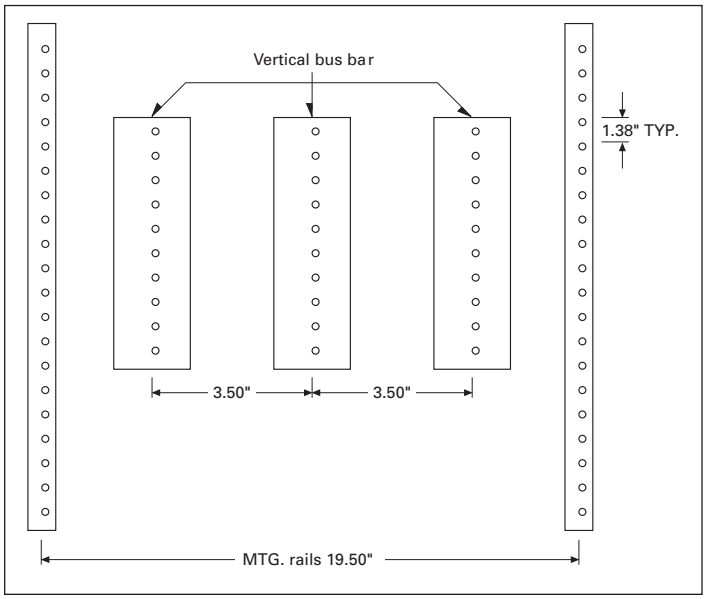


Figure 2. Switch connector and retrofit kits

Note: Dimensions shown apply to CDP and FDP chasis.

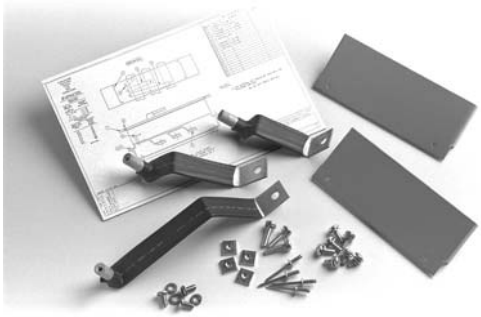
FDP panelboards

FDPW and FDPB connector kits

Each kit includes copper connectors, extension wings (when required), hardware and instructions to mount one FDPW or FDPB switch on a FDP panel chassis.

Connector kit

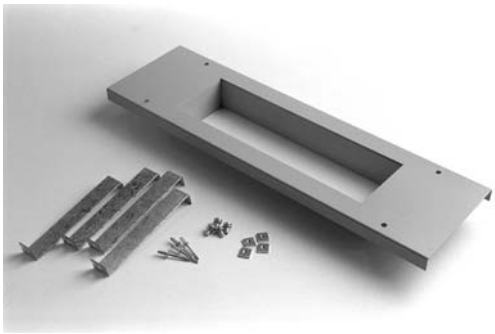
Each kit includes copper connectors, mounting brackets, covers, hardware, and instructions.



Connector kit

"Picture frame"

To permit installation of circuit breakers in a FDP panel, a "picture frame" is required. Add 2X to the breaker size for transition. Contact the Sumter Plant for technical assistance and pricing at 1-800-556-4569.



"Picture frame"

Table 24. FDP panelboards

Space required		Switch ampere rating	Catalog number		
			3-pole FDPW and FDPB		Connector kit
Inches (mm)	X space		240V	600V	
5.50 (139.7)	4X	30–30 60–60 100–100	FDPWT3211R FDPWT3222R FDPWT3233R	FDPWT3611R FDPWT3622R —	KFDP4XT
6.88 (174.8)	5X	100–100	—	FDPWT3633R	KFDP5XT
8.25 (209.6) 12.38 (314.5)	6X 9X	200 400	FDPBS324R FDPW325R	FDPBS364R FDPW365R	KFDPB6X-2 KFDPW9X-4
15.13 (384.3) 15.13 (384.3) 20.63 (524.0)	11X 11X 15X	600 800 1200	FDPW326R FDPW367 FDPW368	FDPW366R FDPW367 FDPW368	KFDPW11X-6 KFDPW11X-8 KFDPW15X-12

Panelboard trim locks

Note: Dimensions in inches (mm).

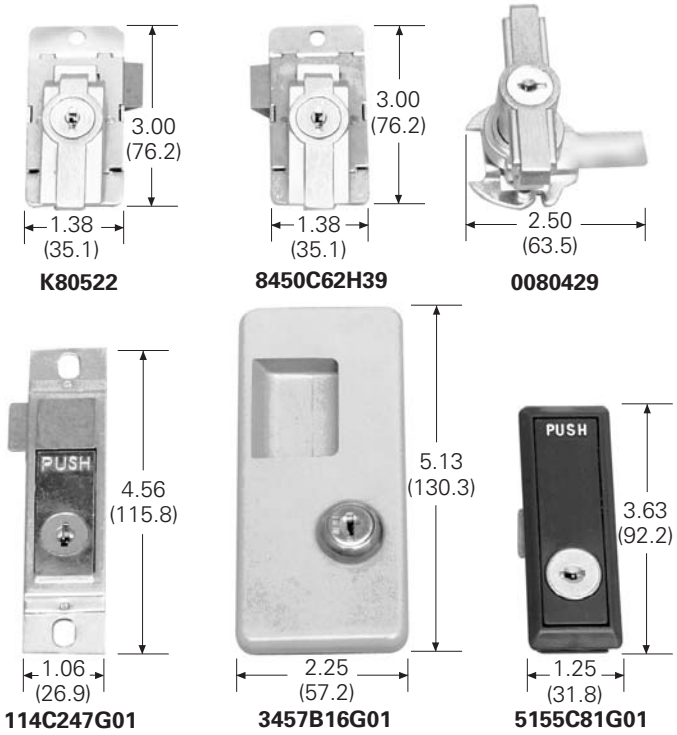


Table 25. Panelboard trim locks

Description	Catalog/style number
For use on left-handed door (hinged on the left side).	K80522
For use on right-handed door (hinged on the right side).	8450C62H39
T handle lock—At one time used on all panels over 48.00 inches (1219.2 mm) in height. Also used on outdoor (N3R) panels.	0080429
Used on lighting and small power panels as standard. Replacement lock uses WEM-3 key.	114C247G01
Used on lighting and small panels as standard.	3457B16G01
This is the present-day lock used on standard PRL1, 2, and 3s. WEM 2 key.	5155C81G01

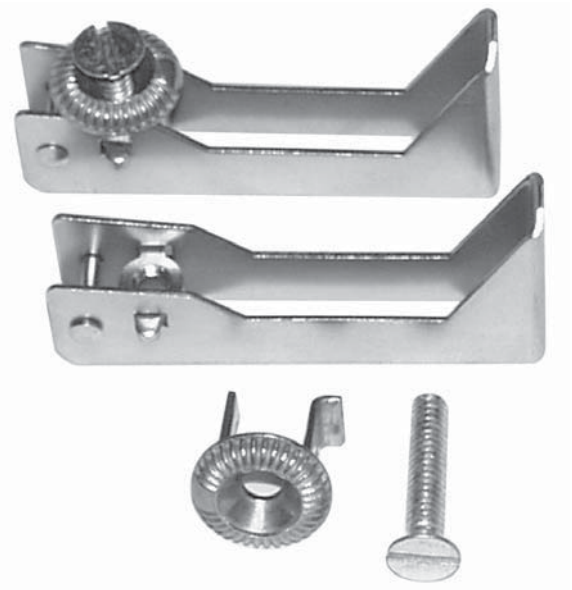
Contact the Sumter Plant Aftermarket Center for prices and details at 1-803-481-6843.

Replacement panelboard trim clamps



414A380G02

- Overall length of 2.50 inches (63.5 mm)
- Used on CDP, FDP, and NQB



7494A11G02

- Overall length of 3.13 inches (79.5 mm)
- Used on B10B and PRL1, 2, 3

These trim clamps are used to mount FASTRIMS. This type of trim used to be supplied as standard on all panels, and is still available today when concealed trim mounting hardware is specified. They come in bags of six. Please note that some larger panels may require more than one bag.

Sumter aftermarket services

All the parts and kits shown in this Renewal Parts data will come from the Sumter Plant Aftermarket Center (order entry location Q49 or SUM).

To extend the life of your existing equipment, the Sumter Aftermarket group will:

- Provide all new components
- Manufacture per original specifications
- Follow UL procedures
- Provide full Cutler-Hammer warranty

Services offered include:

- Parts and kits for CDP and FDP panelboards
- Retrofit kits for WF and WRP distribution sections
- Connector kits for WF and WRP distribution sections
- Emergency service

Additional services

Since virtually all panelboards are supplied to meet specific customer requirements, other parts not listed in this publication might occasionally be needed. Price and availability for parts not shown here may be obtained by contacting the Sumter Plant Aftermarket Center (1-803-481-6843) and providing a complete description of the part, along with the data on the panelboard nameplate.

Should you experience difficulty in determining what replacement parts are needed, contact the Sumter Aftermarket Center, who can provide help to:

- Identify and recommend replacement parts
- Verify the correct connector kits which should be ordered for each circuit breaker or fusible switch
- Provide a recommended spare parts list

⚠ WARNING!

HAZARDOUS VOLTAGE WILL CAUSE SEVERE INJURY OR DEATH. TURN OFF POWER SUPPLY TO EQUIPMENT BEFORE WORKING ON IT.

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Eaton Corporation
Electrical Group
1000 Cherrington Parkway
Moon Township, PA 15108
United States
877-ETN-CARE (877-386-2273)
Eaton.com

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